

Yashwanth Kasanneni

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EDUCATION

NYU Tandon School of Engineering

Master's in Computer Science

Coursework: Analysis of Algorithms, Distributed Systems, Machine Learning, Principles of Database Systems

Expected May 2027

New York, United States

WORK EXPERIENCE

Software Developer Intern

Sept 2023 - Dec 2023

Enabled Analytics

On-Site

- Developed a custom Salesforce-based order management system integrated with WhatsApp, allowing supermarket customers to browse items, place orders, and receive automated responses through a chat-based workflow.
- Built Apex, SOQL/SOSL, and Salesforce Flow automation to process WhatsApp orders in real time, validate item availability, update inventory, prevent duplicate orders, and sync order data across Salesforce records.
- Created Salesforce Lightning interfaces for supermarket staff to monitor WhatsApp orders, update fulfillment status, manage inventory, and access customer/order history from a centralized dashboard, owning the feature end-to-end from design through testing and production deployment while debugging workflow-execution issues across order placement and inventory sync.

PROJECTS

PolicyDiff - AI-Powered Payer Policy Change Detection for Denial Prevention | Python, TypeScript, Next.js, FastAPI, Gemini 2.5 Flash, ClickHouse, Datadog LLM Observability

- Co-built a three-microservice pipeline that monitors healthcare payer policy documents for prior-authorization changes, using SHA-1 diff detection to catch new versions and Gemini 2.5 Flash with a structured-output schema to classify each change — a concrete example of applying an LLM to an internal classification problem rather than a user-facing feature.
- Flagged low-confidence classifications for mandatory human review instead of auto-publishing them, and instrumented every Gemini call through Datadog LLM Observability to trace prompts, outputs, and token usage in production — the operational discipline any LLM-powered system needs to run reliably at scale.

CodeGuard - Agentic Coding Assistant Evaluation and Steering Harness | Python, SWE-bench, Claude Code, GitHub Copilot, Cursor, pylint, bandit

- Built an evaluation harness that runs real-world coding tasks from the public SWE-bench benchmark through multiple agentic coding tools (Claude Code, GitHub Copilot, Cursor), automatically applying each tool's generated patch and scoring it against the benchmark's hidden test suites for functional correctness.
- Layered static analysis (pylint, bandit) on raw pass/fail results to flag style violations and security smells, and iteratively refined task instructions based on failure patterns — improving patch pass rate by 22 percentage points and showing how to systematically evaluate and steer an LLM-based tool rather than trust first-pass output.

CourtIQ - Real-Time Analytics Platform (Lambda Architecture) | Python, Kafka, Spark Structured Streaming, Airflow, Apache Iceberg, PostgreSQL, FastAPI, React

- Built a real-time analytics platform using a Lambda architecture — a Kafka event-streaming speed layer with Spark Structured Streaming alongside an Airflow-orchestrated batch layer writing to Apache Iceberg tables — a distributed-systems pattern for serving both instant and historical views from one event stream.
- Served the merged real-time and batch views through a FastAPI backend and PostgreSQL, ran the full multi-service stack in Docker Compose, and conducted a 24-point architectural gap analysis of the system's failure modes and scaling limits — the kind of systems-design scrutiny that matters once an architecture has to hold up at real scale, not just in a demo.

SecurePR - Secure PR Execution and Verification Platform | Python, FastAPI, Temporal, Kubernetes (GKE), gVisor, Kata Containers, Cilium, Tetragon

- Architected a zero-trust platform that automatically sandboxes and verifies untrusted GitHub PR code end to end — isolating, executing, and monitoring the code, then reporting findings back automatically — the same problem class as safely executing untrusted third-party code without extending its default trust.
- Built the sandbox's isolation layer directly on Linux kernel primitives — using unshare() and fork() to isolate PID, mount, UTS, and IPC namespaces, validated via restricted process visibility and hostname isolation — and began redesigning it to run rootless, replacing sudo with unprivileged user namespaces and UID/GID mapping.

RESEARCH AND PUBLICATIONS

[Effective Analysis of Machine and Deep Learning Methods for Diagnosing Mental Health Using Social Media Conversations](#)

IEEE Transactions on Computational Social Systems

- Analyzed 100K+ tweets and developed NLP pipelines using BERT, RoBERTa, DistilBERT, XLNet, and classical/deep-learning models for behavioral risk-pattern detection.
- Evaluated model accuracy and sentiment features to build predictive systems that identified psychological-condition signals with 97% accuracy.

SKILLS

Languages and Frameworks: Java, Python, SQL, JavaScript/TypeScript, React, Next.js, Node.js, data structures and algorithms

Distributed Systems and Data: Kafka (event streaming), Spark Structured Streaming, PostgreSQL, ClickHouse, microservice architecture

AI Applied to Engineering Systems: LLM integration (Gemini), agentic coding (Claude Code, Copilot, Cursor), LLM observability (Datadog)

Engineering Practices and Security: Static analysis (pylint, bandit), Linux isolation, Kubernetes, Docker, CI/CD

Certifications: Machine Learning Specialization by Andrew Ng, Google Cloud Digital Leader